reasoning can be brought to bear on problems of biological significance. We have found that many of the exercises at the back of each chapter are the jumping off point for the analysis of fascinating biological problems.

Howard, J (2001) Mechanics of Motor Proteins and the Cytoskeleton, Sinauer Associates. Howard's book is a treasure trove of information and concepts.

Nelson, P (2004) Biological Physics: Energy, Information, Life, W. H. Freeman. This book discusses a wide range of topics at the interface between physics and biology.

Benedek, GB, & Villars, FMH (2000) Physics with Illustrative Examples from Medicine and Biology, 2nd ed., Springer-Verlag. This three-volume series should be more widely known. It is full of useful insights into everything ranging from the analysis of the Luria–Delbrück experiment to the action potential.

Sneppen, K, & Zocchi, G (2005) Physics in Molecular Biology, Cambridge University Press. This book presents a series of vignettes that illustrate how physical and mathematical reasoning can provide insights into biological problems.

Jackson, MB (2006) Molecular and Cellular Biophysics, Cambridge University Press. Jackson's book covers many themes in common with our own book and is especially strong on ion channels and electrical properties of cells.

Waigh, TA (2007) Applied Biophysics, Wiley. This interesting book covers many of the same topics covered here.

Harte, J (1988) Consider a Spherical Cow, University Science Books. This book does a beautiful job of showing how order of magnitude estimates can be used to examine complex problems. Those that are intrigued by this book might consider dipping into his Consider a Cylindrical Cow (University Science Books, 2001) as well.

Weinstein, L, & Adam, JA (2008) Guesstimation: Solving the World's Problems on the Back of a Cocktail Napkin, Princeton University Press. This wonderful little book shows by example how to do the kind of estimates advocated here and also is the basis for the rule captured in Figure 1.14.

Goldreich, P, Mahajan, S, & Phinney, S, Order-of-Magnitude Physics: Understanding the World with Dimensional Analysis, Educated Guesswork, and White Lies, online book, available at www.inference.phy.cam.ac.uk/sanjoy/oom/. This excellent work embodies the science of estimation so helpful to biology as well as physics.

Cohen, J (2004) Mathematics is biology's next microscope, only better; Biology is Mathematics' next physics, only better, *PLoS Biology* **2**, e439. This article argues for the way that quantitative reasoning will enrich biology and for the ways that biology will enrich mathematics.

Mahajan, S (2010) Street-Fighting Mathematics, The MIT Press. Much of our mathematics will be of the street-fighting kind and there is nowhere better to get a feeling for this than from Mahajan's book.

1.8 References

Bray, D (2001) Reasoning for results, *Nature* **412**, 863. Source of Darwin's quote at the beginning of the chapter.

Darwin, F (ed.) (1903) More Letters of Charles Darwin: A Record of His Work in a Series of Hitherto Unpublished Letters, Volume I, John Murray.

DiMauro, S, & Schon, EA (2003) Mitochondrial respiratory-chain diseases, *N. Engl. J. Med.* **348**, 2656.

Fawcett, DW (1996) The Cell, Its Organelles and Inclusions: An Atlas of Fine Structure, W. B. Saunders. Fawcett's atlas is full of wonderful electron microscopy images of cells.

Milo, R, Jorgensen, P, Moran, U, et al. (2010) BioNumbers—the database of key numbers in molecular and cell biology, *Nucl. Acids Res.* **38**, D750.

Müller, J, Oehler, S, & Müller-Hill, B (1996) Repression of *lac* promoter as a function of distance, phase and quality of an auxiliary *lac* operator, *J. Mol. Biol.* **257**, 21.

Poustelnikova, E, Pisarev, A, Blagov, M, et al. (2004) A database for management of gene expression data *in situ*, *Bioinformatics* **20**, 2212.

Rubik, BA, & Koshland, DE, Jr. (1978) Potentiation, desensitization, and inversion of response in bacterial sensing of chemical stimuli, *Proc. Natl. Acad. Sci.*, **75**, 2820.

Stent, G (1963) Molecular Biology of Bacterial Viruses, W. H. Freeman.